

Tcpdump

What is Tcpdump?

Tcpdump is a command-line packet analyzer used to capture and inspect network traffic in real time.

Uses:

- Network troubleshooting
 - Security monitoring
 - IDS/IPS analysis
 - Threat hunting
 - Packet capture for Wireshark
-

Installation

Ubuntu/Debian

```
sudo apt update  
sudo apt install tcpdump -y
```

RHEL/CentOS/Rocky Linux

```
sudo yum install tcpdump -y
```

Verify installation:

```
tcpdump --version
```

List Available Interfaces

```
sudo tcpdump -D
```

Example:

```
1.eth0  
2.lo  
3.wlan0
```

Capture All Packets

```
sudo tcpdump
```

or on a specific interface:

```
sudo tcpdump -i eth0
```

Capture Limited Number of Packets

Capture only 10 packets:

```
sudo tcpdump -i eth0 -c 10
```

Display Packet Details

```
sudo tcpdump -vv
```

More verbose:

```
sudo tcpdump -vvv
```

Capture by Protocol

ICMP (Ping)

```
sudo tcpdump icmp
```

Test from another terminal:

```
ping 8.8.8.8
```

TCP

```
sudo tcpdump tcp
```

UDP

```
sudo tcpdump udp
```

Capture by Port

SSH

```
sudo tcpdump port 22
```

HTTP

```
sudo tcpdump port 80
```

HTTPS

```
sudo tcpdump port 443
```

DNS

```
sudo tcpdump port 53
```

Capture by Host

Specific Source/Destination

```
sudo tcpdump host 192.168.1.100
```

Source IP Only

```
sudo tcpdump src host 192.168.1.100
```

Destination IP Only

```
sudo tcpdump dst host 192.168.1.100
```

Capture SSH Traffic from a Specific Host

```
sudo tcpdump host 192.168.1.50 and port 22
```

Save Packets to a File

```
sudo tcpdump -i eth0 -w capture.pcap
```

Stop with:

```
Ctrl + C
```

Read a Saved Capture

```
tcpdump -r capture.pcap
```

Detailed output:

```
tcpdump -nn -r capture.pcap
```

Useful Filters

TCP Port 80

```
sudo tcpdump tcp port 80
```

HTTP and HTTPS

```
sudo tcpdump 'port 80 or port 443'
```

Exclude SSH Traffic

```
sudo tcpdump not port 22
```

Multiple Conditions

```
sudo tcpdump host 192.168.1.10 and port 80
```

Capture SYN Packets

Useful for observing connection attempts:

```
sudo tcpdump 'tcp[tcpflags] & tcp-syn != 0'
```

Monitor SSH Login Attempts

```
sudo tcpdump -i eth0 port 22
```

In another terminal:

```
ssh user@server-ip
```

You will see SSH connection packets.

Capture DNS Queries

```
sudo tcpdump -i eth0 port 53
```

Generate traffic:

```
nslookup google.com
```

Threat Hunting Examples

Detect Excessive Connections from One IP

```
sudo tcpdump -nn
```

Look for repeated packets from the same source IP.

Monitor Failed SSH Activity

```
sudo tcpdump -i eth0 port 22
```

Combine with:

```
sudo tail -f /var/log/auth.log
```

Monitor Web Server Traffic

```
sudo tcpdump -i eth0 port 80 or port 443
```

Common Lab Exercises

Capture Ping Traffic

Terminal 1:

```
sudo tcpdump -i eth0 icmp
```

Terminal 2:

```
ping 8.8.8.8
```

Capture SSH Traffic

Terminal 1:

```
sudo tcpdump -i eth0 port 22
```

Terminal 2:

```
ssh user@192.168.1.10
```

Capture and Save Traffic

```
sudo tcpdump -i eth0 -w lab_capture.pcap
```

Open later in:

Wireshark

```
wireshark lab_capture.pcap
```

What is tcpdump?

Tcpdump is a command-line network packet analyzer used to capture and inspect network traffic.

What is a `.pcap` file?

A packet capture file that stores captured network packets for later analysis.

Which protocol is used for SSH?

TCP port 22.

How do you capture only ICMP packets?

```
sudo tcpdump icmp
```

How do you save packets to a file?

```
sudo tcpdump -w capture.pcap
```

How do you read a saved packet capture?

```
tcpdump -r capture.pcap
```

```
sudo tcpdump -D  
sudo tcpdump -i eth0  
sudo tcpdump icmp  
sudo tcpdump port 22
```

```
sudo tcpdump -i eth0 -w capture.pcap  
tcpdump -r capture.pcap
```